IMPERIAL

Experimental and Computational Investigation of Small-Scale Turbopump Performance with a Novel Barske-Type Impeller

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Abstract

Your abstract goes here. Summarise the key motivations, methodology, and findings of your turbopump FYP.

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Nomenclature

A	Area (m^2)
H	Pump Head (m)
N	Rotational Speed (RPM)
P	Power (W)
Q	Volumetric Flow Rate (m^3/s)
η	Efficiency ($-$)
ρ	Density (kg/m^3)
ω	Angular Velocity (rad/s)

Abbreviations

CFD	Computational Fluid Dynamics
FYP	Final Year Project
$NPSH_a$	Net Positive Suction Head Available
$NPSH_r$	Net Positive Suction Head Required

1 Introduction

Rocket engines require high pressures to operate efficiently. Small scale liquid propulsion such as satellite propulsion and amateur liquid roket propulsion runs almost entirely on pressure fed feed systems due to their simplicity and reliability. Larger orbital rockets use turbopumps because the mass of the required tanks becomes prohibative. However, as space launch becomes exponentially more accessible, there is a growing interest in small scale rocket propulsion systems. For example, lunar landers, satellite propulsion and amateur liquid rockets are seeing a resurgence in interest. Due to the historical focus on large turbopumps for such applications, turbopumps at the smaller scale are not as well documented and have different design considerations. A turbopump is a pump and the turbine that drives it, and both are punished by the small scale.

1.1 Small scale rocket pump challenges

Pump driven propulsion is the solution orbital liquid rockets use to solve the tank mass problem. By storing propellents at lower pressures and pumping them to the required pressures allows for a large reduction in tank mass. The pump is usually driven by turbines powered by burning a portion of the propellents, or by the heat generated by the cooling of the engine. In recent years, battery and electric pump technology has advanced to such a point that for smaller rockets such as the Electron rocket by Rocket Lab [?], electric pumps are starting to become a more viable alternative to turbine driven pumps.

Pumps are generally characterized by their flow rate Q_{opt}, the head H_{opt} and rotor speed n [?]. These three parameters largely decide the pump geometry and type of pump. These three parameters can be related by a quantity known as specific speed, n_q, N_s or ω_s depending on the units and definitions used:

$$n_q = \frac{n\sqrt{Q}}{H^{3/4}} \quad [\mathrm{RPM}, \mathrm{m}^3/\mathrm{s}, \mathrm{m}] \tag{1}$$

Different pump geometries are optimal for different regions of specific speed as can be shown by **??**. Most liquid rocket engines operate between a specifc speed n_q of 20 to 40 and utilize the centrifugal radial impeller to maximize efficiency. This works great for large rocket engines, but scaling this down to smaller engines becomes increasingly difficult. All rocket engines require high pressures, but smaller rocket engines require less flow rate. To stay in the specific speed according to **??** that allows for high efficiency in the centrifugal pump requires increasing rotational speed.

To achieve the specific speed of 20 for a smaller rocket engine of around 5 kN requiring 1.0 kg/s of kerosene at 50 bar leads to rotational speeds of around 71 051 rpm. Such high rotational speeds leads to challenging mechanical design conditions. To solve this problem, a single pump can be separated into multiple stages, separating the head rise to increase the specific speed of each stage. This is a common solution to pumping hydrogen. This thesis however addresses the other solution, which is to deal with the lower specific speed.

As shown in **??**, at lower specific speeds, the partial emission pump becomes the optimum efficiency pump geometry at specific speeds of 5-10. However, even though these pumps are the most efficient at these low specific speeds, they physically cannot achieve the high efficiencies of a full emission pump. This is the main limitation of trying to operate at low specific speeds.

Due to its lower efficiency and the lack of pump fed liquid rockets for small engines, there is a lack of literature, design knowledge and experimental data surrounding these pumps. This thesis aims to address this gap by reviewing existing literature, designing and manufacturing a partial emission centrifugal pump and comparing its experimental performance specifically in the use case of rocket engines.

1.2 Small scale rocket turbine challenges

On the other end of the turbopump shaft is the turbine which drives the pump. The turbine faces the same problem as the pump. If using the gas generator cycle, the small turbine leads to lower blade speeds U. To extract the maximum work from from the gas, gas velocity C_0 must be high, leading to a supersonic blade turbine. The high gas velocity relative to blade speed leads to the turbine running at a lower efficiency than the optimum $U/C_0 \approx 0.5$ [?]. The smaller mass flow also cannot fill the annulus at a sensible blade height, leading to partial admission where only a fraction of the annulus has gas flowing through it at any given time. This makes the stage lossy and awkward to design. In addition to these design limitations, supersonic turbine design is far less studied than subsonic, where methods are scattered across older sources.

This thesis tackles the design of a turbopump in the context of small scale liquid rocket engines which forces both pump and turbine off optimum design conditions, then demonstrates their coupled performance in a full integration test. The thesis is framed around the design of a 20 000 rpm turbopump using water as a inert simulant for the pumped rocket propellents. Then, a supersonic partial admission impulse turbine is designed to drive the pump using compressed air as the working fluid. The two are then coupled together and tested as a single system to evaluate performance.

1.3 Aims and Objectives

The aim of this thesis is to investigate the performance of a small scale rocket low specific speed turbopump with a Barske pump and a supersonic partial admission turbine. To this end, the objectives of this thesis are:

1. Pump characterisation: Determine the head-flow curve, efficiency, and cavitation behaviour of a Barske-type impeller across operating speeds
2. Turbine design and model validation: Design a single-stage partial-admission supersonic impulse turbine with a first-principles computational toolchain, and validate those models against the achievable test data.
3. Turbopump integration: Demonstrate coupled turbopump operation at scaled off-design conditions and perform analytical extrapolation to the full 20 000 rpm gas generator regime.
4. Computational toolchain assessment: Assess the 1D/analytical design toolchain against the experimental data, and establish where analytical prediction is sufficient for this class of machine.

1.4 Scope and Limitations

This thesis covers the theory, design, manufacture and testing of a small scale turbopump system on inert propellents. The pump was tested up to 9000 rpm, significantly lower than the design speed of 20 000 rpm in the first and final iterations of the conducted experiments. This means the data validation can only happen for the off-design point and extrapolation to the design point is required. Cryogenic and hot gas design and testing are strictly outside the scope of tis thesis. Operational safety procedures are included in the appendix.

1.5 Thesis Structure

Chapter 2 establishes the pump theoretical background: pump specific speed, Barske impeller characteristics, impulse turbine velocity triangle theory, and the relevant prior literature. Chapter 3 details the pump design and sizing methodology. Chapter 4 covers turbine design, including structural analysis of the rotor at 20 000 rpm. Chapter 5 describes the custom instrumentation and data acquisition system. Chapter 6 documents the test rig and assembly. Chapter 7 presents the experimental results. Chapter 8 discusses the findings in the context of design predictions and prior literature. Chapter 9 draws conclusions and recommends future work.

2 Design Background

2.1 Barske Impeller

Centrifugal pumpss have multiple main sources of losses. The one that becomes significant at low specific speeds is the disc friction losses. When a rotor object rotates in a fluid, the shear stresses in the fluid cause a drag torque on the object. In most centrifugal pumps the impeller is disc shaped, and the loss is called disc friction loss. . With low specific speed pumps, disk frction is the main source of loss. At $n_q = 10$, $\psi_{\mathrm{opt}=1}$ and $\mathrm{Re} = 10^7$, the disc friction is approxiomately 30% of the usefulf power of the impeller [?]. Gulich suggests that a high head coefficient reduces the diameter and consequently less disk frciont since $P_{RR} \sim D^5$. This is the reason why low specific speed centrifugal pumps physically cannot achieve the same high efficiencies as higher specific speed pumps.

On the other hand, at low specific speeds, impeller losses are low as shown by **??**. As a result, surface finish on the impeller blades are largely unimportant, allowing for simpler manufacturing processes. Conversely, collector losses become dominant. Gulich explains [?] that for $n_q < 12$ even higher efficiencies are expected than with volutes. This simplifies the design and manufacturing of the pump, which is a significant advantage at small scales where tolerances are both harder to achieve but contribute to more significant losses. This is important as the high fluid velocity at the peripheri makes the surface finish of the annular casing of primary importance for pump efficiency [?].

The Barske impeller pump design which was developed for the need for manufacturing simplicity and low specific speed applications became one of the highest efficiency designs at this specific speed. The barske impeller operates differently to a normal full emission pump. In a conventional centrifugal pump, impeller blade geometry aims to perfectly guide fluid flow along the blade surfaces in a smooth spiral streamline out of the impeller into the volute [?]. The barske impeller is instead designed to induced almost a pure forced vortex flow where $u = \omega r$ and the fluid is essentially rotating as a solid body with the impeller [?]. This also explains the unimportance of surface finish on the blades, as low relative velocity between the fluid and the impeller means lower friction losses.

Barske comments that the number of impeller blades does not seem to be critical, with 3-6 being operated successfully. However, altering the number of blades can be used to elminite fluid vibrations [?]. Lock provides corrections for blades from 1 to 16 [?].

The open impeller choice is to further reduce disc friction losses by minimizing surface area. Conventional impellers uses shrouds to guide the flow and reduce secondary flows. However, as in a Barske pump the impeller serves only to generate the forced vortex flow, this was deemed unnecessary by at least Barske and Lock. This also removes axial thrust loads on the impeller, helping with sealing and mechanical design of the shaft. Also, simplify manufacturing.

Conventional pump design with current design tools produce nonsensical geometries and results at low specific speeds because of one number that is used in the design process, which is the slip coefficient. The slip coefficient determines the angular deflection of the fluid leaving the impeller from the blade angle at the impeller exit. This number is usually determined empirically based on regression fits from existing data. However, at $n_q \approx 7$, the equations relating blade exit angle, slip coefficient and head coefficient mathematically does not have a converging root. Physically, there is no blade geometry that can bend the blade enough to keep the flow attached to the blade. This is further motivates the forced vortex Barske architecture, as it does not rely on attached flow, removing slip factor from the design process entirely.

All theories relating to pump and turbine design can be derived from the conservation of angular momentum by Newton's second law of motion. In turbomachinery, this is usually expressed in terms of the Euler turbine equation:

$$P = T_{shaft}\omega = \rho Q \left(r_2 c_{2u} - r_1 c_{1u}\right)\omega = \rho Q \left(u_2 c_{2u} - u_1 c_{1u}\right) \tag{2}$$

Rewriting this equation in terms of specific work $Y = P/\rho Q$ per unit mass:

$$Y = \left(r_2 c_{2u} - r_1 c_{1u}\right)\omega = u_2 c_{2u} - u_1 c_{1u} \tag{3}$$

derive from bernouli, with rotating frame substitution $c^2 = w^2 - u^2 + 2 \times u \times c_u$ and neglecting compressibility losses:

$$p_1 + \frac{\rho}{2}w_1^2 - \frac{\rho}{2}u_1^2 = p_2 + \frac{\rho}{2}w_2^2 - \frac{\rho}{2}u_2^2$$

and as such the theoretical rise in static pressure:

$$\Delta p_{st} = p_2 - p_1 = \frac{\rho}{2}\left(w_1^2 - w_2^2\right) + \frac{\rho}{2}\left(u_2^2 - u_1^2\right) \tag{4}$$

In Barske, Lobanoff and Lock the relative velocity w is assumed to be negligable, ending up with:

$$\Delta p_{st} = \frac{\rho}{2}\left(u_2^2 - u_1^2\right) \tag{5}$$

Then, the kinetic energy increase assuming negligible inlet prerotation is:

$$\Delta p_{st} = \frac{\rho}{2}u_2^2 \tag{6}$$

?? is the theoretical total pressure a barske impeller generates at the blade tips. In a barske pump, the diffuser is responsible for converting this kinetic energy into static pressure by slowing down the fluid in an expanding diffuser. Barske introduces a loss coefficient for the kinetic energy which can be recovered through the diffuser to be between 0.2 and 0.5. Barske uses the symbol ψ for this coefficient, but as it conflicts with the commonly used symbol for head coefficient, this paper will instead use c_k. Combining the static and kinetic pressure rise, the total increase in pressure is:

$$p_t = p_{s_{th}} + c_k p_{d_{th}} = \frac{1}{2}\rho\left((1 + c_k)u_2^2 - u_1^2\right) \tag{7}$$

This forms a simple basic model for barske pump performance.

2.2 Barske Pump sizing

The barske pump is sized by the methods provided by barske [**?**] with the assitance of Gulich. First the inlet is sized by choosing a reasonable inlet velocity and calculating the inlet area:

$$d_0 = \sqrt{\frac{4\dot{m}}{\pi\rho c_0}} \tag{8}$$

The impeller blade inlet diameter is enlarged by a small amount as some pre-rotation is desirable for increased head. For example, $d_1 \sim 1.1 d_0$.

By substituting $u = \pi n d/60$ into **??**, the required impeller tip diameter can be calculated:

$$d_2 = \sqrt{\frac{1}{1 + c_k}\left(\frac{7200 p_t}{\rho n^2 \pi^2} + d_1^2\right)} \tag{9}$$

Barske then gives these guidelines for other dimensions of the pump:

$$s_{ax} \leq \min\left(0.75, \frac{d_2}{100}\right) \tag{10}$$

$$b_1 \geq \frac{1}{4}d_1 \tag{11}$$

$$b_2 \geq \max\left(b_1\frac{d_1}{d_2}, 3s_{ax}\right) \tag{12}$$

$$\tag{13}$$

For the annular casing sizing, Barske doesn't have very specific numbers, so Gulich's guideline where B/H = 1.5 to 2.5 can be followed:

$$B = 2s_a x + b_2 \tag{14}$$
$$H = B/2 \tag{15}$$

To size the diffuser, Barske suggests sizing the diffuser throat to be around 0.8 to 0.9 of u_2, with the area of the diffuser outlet to be 3-4 times the area of the diffuser throat.

2.3 Barske Pump efficiency

Barske assumes that the disc friction losses can be approximated as a circular disc in a rotating fluid. Converting the equation into metric units gives:

$$P_{RR}[\mathrm{W}] = 1956 \times \rho\nu^{0.2} \left(\frac{n}{1000}\right)^{2.8} \left(d_2^{4.6} + 4.6 d_1^{3.6} b_1\right) \tag{16}$$

As the hydraulic efficiency is defined as the useful power out of the pump over the total power into the pump, efficiency can be defined as:

$$\eta_h = \frac{\text{Useful power}}{\text{Input power}} = \frac{P}{P_{th} + P_{RR}} = \frac{Qp}{Qp_{th} + P_{RR}} = \frac{p}{p_{th} + P_{RR}/Q} \tag{17}$$

If total shaft efficiency is desired, the mechanical losses can be included:

$$\eta = \frac{p}{p_{th} + \frac{P_F + P_{Mech}}{Q}} \tag{18}$$

Lock provides a refined model which takes into account prerotation, nozzle losses and correction coefficients taking into account theoretical two dimensional flow solutions by A. Busemann [?].

By writing the total theoretical pressure gain in head:

$$H_T = \frac{U_2^2 - U_1^2}{2g} + \frac{U_2^2}{2g} \tag{19}$$

Lock uses busemann's corrections and derives a refined total head, taking into account prerotation:

$$H_{TH} = h_0 \frac{U_2^2}{g} - C_H\, K_1 \left(1 - \frac{Q}{Q_{op}}\right) \frac{D_1}{D_2} \frac{U_2^2}{g} \tag{20}$$

$$H_{SP} = H_{TH} - \underbrace{\zeta_{LD} \left[1 - \left(\frac{D_3}{D_4}\right)^2\right]^2 \frac{V_3^2}{2g}}_{\text{diffuser-nozzle loss}} - \frac{V_4^2}{2g} + \frac{V_0^2}{2g} \tag{21}$$

Where $\zeta_{LD} = 0.194$ is the diffuser loss coefficient from Hydraulic Institute's Pipe Friction Manual [?].

By expressing $V^2/2g$ by $(8/g\pi^2)Q^2/D^4$ with $V = 4Q/\pi D^2$, the head can be expressed as a function of flow rate:

$$H_{SP} = h_0 \frac{U_2^2}{g} - C_H\, K_1 \left(1 - \frac{Q}{Q_{op}}\right) \frac{D_1}{D_2} \frac{U_2^2}{g} - \frac{8Q^2}{g\pi^2} \left(\frac{\zeta_{LD}}{D_3^4} + \frac{1}{D_4^4} - \frac{1}{D_0^4}\right) \tag{22}$$

To find the optimum flow rate used to shape the pump curve, the derivative of the head with respect to flow rate can be taken and set to zero:

$$Q_{op} = \sqrt{\frac{h_0 \frac{U_2^2}{g}}{\frac{24}{g\pi^2} \left(\frac{\zeta_{LD}}{D_3^4} \left(1 - \left(\frac{D_3}{D_4}\right)^2\right) + \frac{1}{D_4^4} - \frac{1}{D_0^4}\right)}} \tag{23}$$

Lock and Barske suggest that the flow cutoff point is limited by the diffuser throat. To find the static pressure at the diffuser throat, reversing the transfer between dynamic head into static head provided by the previous equations :

$$H_3 = H_{SP} + \frac{8Q^2}{g\pi^2} \left(\frac{\zeta_{LD}}{D_3^4} \left(1 - \left(\frac{D_3}{D_4}\right)^2\right) + \frac{1}{D_4^4} - \frac{1}{D_3^4}\right) \tag{24}$$

Then solving for the Q at which H_3 is equal to vapor pressure head $H_{vap} = \frac{1}{\rho g}\left(p_{vap} - p_{inlet}\right)$ gives the flow cutoff point.

2.4 Impulse Turbine

Turbines work with the same Euler turbine equation. In the supersonic impulse turbine design, the blades are only occupy a small outer portion of the blisk, so the turbine equation can be simplified:

$$\Delta h_{\text{use}} = c_{3u}u_3 - c_{4u}u_4 = u(c_{3u} - c_{4u}) \tag{25}$$

From **??**, it is understood that a higher flow velocity difference $c_{3u} - c_{4u}$ per unit mass will lead to a higher Δh_0 according to equation **??**. This means that high absolute velocities, and for the maximum amount of the flow angled in the circumferential direction is optimal. For this reason, gas generator cycle turbines tend to take advantage of the high pressure ratios and thus the supersonic velocities. Then for the angles, the highest turning angle achievable is optimal. The amount of flow turning is usually limited by the achievable geometry.

Leading to the ideal efficiency of an impulse turbine to be:

$$\eta_{\text{ideal}} = 4\phi(\cos\beta_3 - \phi) \tag{26}$$

Where ϕ is the flow coefficient defined as $\phi = c_3/u$ and β_3 is the blade angle at the inlet. The maximum ideal efficiency is achieved at $\phi = \cos\beta_3/2$. By taking the derivative of **??**, the ideal U/C_0 is $0.5\cos\beta_3$. For a small turbine with a diameter of

NASA special publication 8110 shows that for low U/C_0 turbines, impulse turbines are more efficient, even if reaction turbines can have a higher overall efficiency, just not at the low U/C_0 regime. In addition, the same conclusion is drawn by comparing the influence of tip clearance on the two types, where impulse turbines are more robust to tip clearance. This is especially important for small scale turbines where manufacturing tolerances are more difficult to achieve and contribute to more significant losses [**?**].

Another problem with designing small turbines is with the low blade height required. By calculating the mass flux through the turbine, the blade height can be designed:

$$H_3 = \frac{\dot{m}}{\rho_3 c_{3m} d_{\text{mean}}} \tag{27}$$

Where ρ_3 is the density of the turbine gas at the turbine inlet. Using the example turbine above, the required blade height would be . By introducing partial admission, the mass flow through the turbine can be restricted to a smaller arc around the annulus at a slight cost to efficiency. The partial admission ratio ζ can be included into the blade height formula as so:

$$H_3 = \frac{\dot{m}}{\zeta \rho_3 c_{3m} d_{\text{mean}}} \tag{28}$$

By reducing the partial admission to , the blade height is increased to . The design of the turbine blade height is very important due to multiple reasons. Firstly, the aspect ratio of the turbine blade determines the magnitude of of the wall boundary layer blockage. For this reason, a aspect ratio between is desired.

Another way to increase the blade height is to decrease the meridonal velocity c_{3m}. This is achieved by increasing the angle that the flow has to turn as $c_{3m} = c_3\sin(\beta_3)$.

For supersonic flow to develop within a turbine blade passage, a condition called "starting" must be satisfied. On the startup of a supersonic turbine, a normal shock starts at the inlet of the blade passage and as to pass through the whole passage before the flow into and out of the turbine passage can be supersonic.

2.5 Method of Characteristics Blade Profile Design

The design so far is based on 1D mean line theory. The design of the actual blade profile which will turn the supersonic flow in the desired direction is often accomplished using an application of the Method of Characteristics to transition a uniform flow field into vortex flow. .

2.5.1 Supersonic Turbine Blade Terminology

Geometry is separated into the turbine blade, and the passage between the flow. Two surfaces are separately designed, and it is common to refer to the top surface as the convex, and the bottom surface as the concave one. This is in relation to the solid blades themselves as opposed to an alternate designation by the top and bottom of the blade passage.

Vortex flow is a special type of supersonic flow in which flow follows a theoretical perfect arc around a central point. This is a very special type of flow field that is achieved by conditions where expansion and compression waves cancel each other to allow shockless rotation of the flow field.

2.5.2 Finite leading edge

In the theoretical method of characteristics, both the upper and lower surface at the turbine inlet is perfectly parallel to the nozzle exhaust angle. This is problematic because this implies the two surfaces will end at a infinitely sharp point where the at the infinitesimal leading edge, the angle between the upper and lower surfaces is zero. This is not manufacturable

- turbine discussion can continue, first type wedge angle, MOC, supersonic starting, radius, boundary layers. - in either design section I can talk about efficiencies - from MOC, talk about upper and lower surface mach numbers, which have the NASA requirements, but also separation requirements. Note that upper surface separation is less important

3 Pump Design

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References

A First Appendix Title

Appendix content goes here (e.g., calibration curves, large tables, or code).